

Trial 1: Questions for Chatbot.

06 Jun 2024

Available Documents:

 GB 4785-2019 Lighting installation -EN.pdf	3/28/2024 5:00 PM	Microsoft Edge PDF Document	3,967 KB
 GB 25991-2010 Automotive Headlamps with LED Light Sources and/or LE...	3/25/2024 10:34 AM	Microsoft Edge PDF Document	210 KB
 GB T 26773-2011 Intelligent Transport Systems Lane Departure Warning ...	3/25/2024 10:39 AM	Microsoft Edge PDF Document	308 KB
 GB T 38186-2019_advanced emergency braking-en.pdf	3/25/2024 10:38 AM	Microsoft Edge PDF Document	284 KB
 UN ECE R123.pdf	3/25/2024 10:38 AM	Microsoft Edge PDF Document	3,147 KB
 UN ECE R130.pdf	3/25/2024 10:39 AM	Microsoft Edge PDF Document	628 KB
 UN ECE R131.pdf	3/25/2024 10:38 AM	Microsoft Edge PDF Document	435 KB

Sr No.	Document	Questions	Answers	Feedback	Reviewer (Owner of Question)
1	GB 4785	Whats the difference between Grouped and Combined lamps?	According to the UN Regulation No. 48 on vehicle lighting installation, there is a distinction made between "grouped lamps" and "combined lamps": Grouped lamps (paragraph 2.7.4) are devices that have separate apparent surfaces in the direction of the reference axis, separate light sources, but a common lamp body. Combined lamps (paragraph 2.7.5) are devices that have separate apparent surfaces in the direction of the reference axis, but a common light source and a common lamp body. In simpler terms: Grouped lamps have separate light sources (bulbs/LEDs) but share the same outer housing/body. Combined lamps have a single shared light source, but the light is projected through separate lenses/optics to produce separate apparent light surfaces. So the key difference is whether the lamps share a common light source (combined) or have separate light sources (grouped), while still being integrated into a single housing unit. This distinction is important for determining the photometric requirements and visibility angles that apply to each lamp function when they are grouped or combined together.	Correct	
		Can dipped-beam headlamp and main-beam headlamp for front lighting system?	Based on the text in the regulation, yes, a vehicle can have both dipped-beam headlamps (also called passing-beam headlamps) and main-beam headlamps as part of its front lighting system. Specifically: Section 6.1 covers the requirements for "Main-beam headlamps" and states they are mandatory on motor vehicles but prohibited on trailers. Section 6.2 covers the requirements for "Dipped-beam headlamps" (passing-beam headlamps) and states they are also mandatory on motor vehicles but prohibited on trailers. So both dipped-beam and main-beam headlamps are required front lighting devices for motor vehicles according to this UN Regulation No. 48 on installation of lighting and light-signaling devices. The dipped-beam provides a lower intensity beam for normal driving to avoid dazzling oncoming traffic, while the main-beam provides a higher intensity beam for better illumination when there is no oncoming traffic.	Correct	
		what is color of End Outline marker lamp?	According to paragraph 5.15 of the regulation, the color requirements for End-outline marker lamps are: White in front, red at the rear. So the end-outline marker lamps should be white in color at the front of the vehicle, and red in color at the rear of the vehicle.	Correct	
		Can yellow lamp used as front fog lamp?	According to paragraph 5.15 of UN Regulation No. 48, the permitted colors for front fog lamps are: "Front fog lamp: White or selective yellow" So yes, a selective yellow lamp can be used as a front fog lamp according to this regulation.	Correct	

		Can red color light placed in the front of the vehicle?	According to paragraph 5.10. of UN Regulation No. 48, no red light which could give rise to confusion shall be emitted from a lamp towards the front of the vehicle. Specifically: 5.10. No red light which could give rise to confusion shall be emitted from a lamp as defined in paragraph 2.7. in a forward direction and no white light which could give rise to confusion, shall be emitted from a lamp as defined in paragraph 2.7. in a rearward direction. So red lights are generally prohibited from being placed in the front of the vehicle, as they could be confused with a rear lamp like a stop lamp or rear position lamp. The regulation aims to prevent potential confusion from improper color lighting towards the front or rear of the vehicle. The only exceptions mentioned are that this requirement shall not apply to lighting devices fitted for the interior lighting of the vehicle, and in case of doubt, there are specific zones defined in paragraphs 5.10.1 and 5.10.2 to check if the red light towards the front would be directly visible and potentially cause confusion. But in general, to comply with UN R48, red lights should not be installed towards the front of the vehicle where they could be confused with rear lamps by other drivers.	Correct	
		Can white light can be placed at the back of the vehicle?	According to paragraph 5.15 of UN Regulation No. 48, white light is not permitted to be emitted from lamps at the rear of the vehicle, except for reversing lamps and rear registration plate lamps. Specifically, it states: "5.15. The colours of the light emitted by the lamps are the following: ... Reversing lamp: White Rear registration plate lamp: White ... No white light which could give rise to confusion, shall be emitted from a lamp as defined in paragraph 2.7. in a rearward direction." So white light is only allowed at the rear for reversing lamps and rear registration plate lamps. For other rear lamps like rear position lamps, stop lamps, rear fog lamps etc., white light is not permitted and red light must be used instead to avoid confusion	Perfect	
		What are 1,1a,1b,2a, 2b,5,6 in direction indicator lamps?	The numbers 1, 1a, 1b, 2a, 2b, 5, 6 refer to different categories of direction indicator lamps based on their arrangement and positioning on the vehicle, as defined in UN Regulation No. 48. Specifically: 1, 1a, 1b - Categories for front direction indicator lamps 1 - If the distance between the edge of the apparent surface and the dipped-beam headlamp is 40 mm 1a - If the distance is > 20 mm but < 40 mm 1b - If the distance is 20 mm 2a, 2b - Categories for rear direction indicator lamps 5, 6 - Categories for side direction indicator lamps 5 - For M1 vehicles and vehicles 6m length 6 - For vehicles > 6m length The different categories account for the positioning of the direction indicators relative to the dipped beam headlamps (for front) and the overall vehicle length (for side), to ensure proper visibility requirements are met for each configuration.	Missing Information	
		is cornering lamp mandatory?	Based on the text in the UN Regulation No. 48, cornering lamps are optional for motor vehicles, not mandatory. Specifically, paragraph 6.20.1 states: "6.20.1. Presence Optional on motor vehicles." So vehicle manufacturers can choose whether or not to install cornering lamps on their vehicles. The regulation provides the requirements for cornering lamps if a manufacturer decides to install them, but does not make them mandatory equipment.	Correct	
		does reflective tape come under light and light signalling?	Based on the UN Regulation No. 48 on the installation of lighting and light-signaling devices for vehicles, reflective tape or conspicuity markings are covered under this regulation. Specifically, Section 6.21 of the regulation deals with "Conspicuity markings" which includes requirements for reflective markings on certain categories of vehicles like trucks and trailers. Some key points regarding conspicuity markings from the regulation: - It defines "conspicuity marking" as a device intended to increase the conspicuity of a vehicle, when viewed from the side or rear, by the reflection of light emanating from a light source not connected to the vehicle. - It specifies the presence, arrangement, position, visibility requirements for conspicuity markings like contour markings, partial contour markings and line markings on different vehicle categories. - It allows the use of retro-reflective devices like retro-reflectors or brackets containing retro-reflecting material compliant with the photometric requirements, as part of the required conspicuity markings. So in summary, while reflective tape itself is not a lighting or light-signaling device, the UN Regulation No. 48 does cover the installation requirements for conspicuity markings which can include the use of retro-reflective materials like reflective tapes on certain vehicle categories to increase their visibility.	Correct	
		standard weight of a person for testing?	According to Annex 5 of UN Regulation No. 48, the mass of passengers shall be calculated on the basis of 75 kg per person for determining the loading conditions for testing the variation in vertical orientation of dipped-beam headlamps. Specifically, Annex 5, paragraph 1 states: "For the following tests, the mass of the passengers shall be calculated on the basis of 75 kg per person." So the standard weight assumed for each person/passenger is 75 kg when testing the effects of vehicle loading on headlamp aim and orientation.	Correct	

		can dipped beam uses as a main beam?	Based on the UN Regulation No. 48 for vehicle lighting installation, the dipped-beam headlamps (also called passing-beam headlamps) cannot be used as the main-beam headlamps. The key points are: 1. The regulation defines main-beam headlamps (paragraph 2.7.9) and dipped-beam headlamps (paragraph 2.7.10) as separate lighting devices with different functions. 2. Paragraph 6.2.7.2 states that the dipped-beam may remain switched on at the same time as the main beams, implying they are separate beams. 3. Paragraph 6.2.7.1 requires that when switching from dipped-beam to main-beam, the main-beam headlamps must be switched on, indicating the main-beam cannot be produced by the dipped-beam lamps alone. 4. The electrical connections in paragraph 6.2.7.1 specify that the control for changing over to the dipped-beam shall switch off all main-beam headlamps simultaneously. So in summary, while the dipped-beam can remain on together with the main-beam, the dipped-beam headlamps alone cannot be used to provide the main high-beam function according to this regulation. Separate dedicated main-beam headlamps are required.	Correct	
		what are the light functions to be kept rear of the vehicle?	According to the regulation, the following lighting and light-signalling devices are required to be installed at the rear of the vehicle: Mandatory: - Rear position lamps (6.10) - Stop lamps (6.7) - Rear retro-reflectors, non-triangular (6.14) - Rear retro-reflectors, triangular (for trailers only) (6.15) - Rear registration plate illuminating device (6.8) Optional: - Reversing lamps (6.4) - Rear fog lamps (6.11) - Rear direction-indicator lamps (6.5) - End-outline marker lamps (6.13) - Conspicuity markings (6.21) - Emergency stop signal (6.23) - Manoeuvring lamps (6.26) The main rear lighting functions that must be kept are the rear position lamps, stop lamps, and rear retro-reflectors to provide rear visibility and signal braking. The other lamps listed are either mandatory for certain vehicle categories or optional additional lighting devices	Mistakes Rear for lamp should be mandatory	
		What lamp should be fitted for passenger vehicles?	According to the UN Regulation No. 48 on the installation of lighting and light-signalling devices for motor vehicles, the following lamps are mandatory for passenger vehicles (categories M1 and M2): 1. Main-beam headlamps (6.1.1) 2. Dipped-beam headlamps (6.2.1) 3. Front direction-indicator lamps (6.5.1) 4. Rear direction-indicator lamps (6.5.1) 5. Hazard warning signal lamps (6.6.1) 6. Stop lamps (6.7.1) 7. Rear registration plate illuminating device (6.8.1) 8. Front position lamps (6.9.1) 9. Rear position lamps (6.10.1) 10. Rear retro-reflectors, non-triangular (6.14.1) 11. Side retro-reflectors, non-triangular (6.17.1 for vehicles over 6m length) 12. Daytime running lamps (6.19.1) Some other lamps like front fog lamps, rear fog lamps, reversing lamps, end-outline marker lamps, side marker lamps are optional for passenger vehicles according to this regulation. The presence, number, color, position and other requirements for each of these mandatory and optional lamps are specified in detail in the corresponding sections of the regulation	M3 is missing	